

PHYS 201: Classical Mechanics

Lecture 4

Newton's Laws of Motion and Energy Conservation

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1. Newton's Three Laws of Motion

First Law (Law of Inertia)

An object at rest stays at rest, and an object in motion stays in motion with the same speed and in the same direction, unless acted upon by a net external force.

Second Law

The net force acting on an object is equal to the rate of change of its momentum. For constant mass:

$$F = ma$$

where F is the net force (in Newtons), m is the mass (in kg), and a is the acceleration (in m/s^2).

Third Law

For every action, there is an equal and opposite reaction. If object A exerts a force on object B, then object B exerts an equal and opposite force on object A.

$$F_{AB} = -F_{BA}$$

2. Work and Kinetic Energy

Work Done by a Constant Force

Work is defined as the dot product of force and displacement:

$$W = \mathbf{F} \cdot \mathbf{d} = Fd \cos(\theta)$$

where θ is the angle between the force vector and the displacement vector. Work is measured in Joules (J).

Kinetic Energy

The kinetic energy of an object of mass m moving with speed v is:

$$KE = \frac{1}{2} m v^2$$

Work-Energy Theorem

The net work done on an object equals the change in its kinetic energy:

$$W_{\text{net}} = \Delta KE = KE_f - KE_i$$

3. Potential Energy and Conservation of Energy

Gravitational Potential Energy

Near Earth's surface, the gravitational potential energy of an object at height h is:

$$PE = m g h$$

where g is approximately 9.8 m/s^2 .

Elastic Potential Energy

For an ideal spring obeying Hooke's Law, the potential energy stored is:

$$PE_{\text{spring}} = (1/2) k x^2$$

where k is the spring constant and x is the displacement from equilibrium.

Conservation of Mechanical Energy

In the absence of non-conservative forces (like friction or air resistance), the total mechanical energy of a system remains constant:

$$KE_i + PE_i = KE_f + PE_f$$

Example: A ball dropped from height h

A ball of mass m is dropped from rest at height h . Just before hitting the ground, all potential energy has converted to kinetic energy:

$$m g h = (1/2) m v^2 \Rightarrow v = \sqrt{2gh}$$

4. Momentum and Collisions

Linear Momentum

Momentum is defined as:

$$p = m v$$

Conservation of Momentum

In an isolated system with no external forces, total momentum before a collision equals total momentum after the collision:

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

Elastic vs. Inelastic Collisions

In an **elastic collision**, both momentum and kinetic energy are conserved. In a **perfectly inelastic collision**, the objects stick together after impact, momentum is conserved, but kinetic energy is not (some is converted to heat, sound, or deformation).

Summary Table

- Newton's 2nd Law: $F = ma$
- Kinetic Energy: $KE = (1/2) m v^2$
- Gravitational PE: $PE = m g h$
- Spring PE: $PE = (1/2) k x^2$
- Momentum: $p = m v$
- Elastic collisions conserve both momentum AND kinetic energy
- Inelastic collisions conserve momentum ONLY